

Assignment: Hive Practice using Airline Datasets

Part A: Table Creation and Data Loading

1. Create a managed table named airports with appropriate columns and data types based on the dataset provided.
 2. Create a managed table named airlines with the correct schema and data types.
 3. Create a managed table named routes using the dataset structure provided.
 4. Load data from local files (airports.txt, airlines.txt, routes.txt) into their respective tables.
 5. Create an external table for airports that references data stored in a specific HDFS directory. Explain the difference in behavior when dropping the internal vs. external table.
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Part B: Joins and Querying

6. Display all distinct countries that have at least one active airline operating.
 7. List all airports along with their corresponding airline names by performing an appropriate join.
 8. Find all routes that operate between airports located in the same country.
 9. Retrieve details of airlines that have more than 10 routes originating from Indian airports.
 10. Display the top 5 countries with the highest number of airports.
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Part C: ORC and Parquet Storage Formats

11. Create a copy of the routes table in ORC format.
12. Create another copy of the routes table in Parquet format.
13. Compare the storage size of the ORC and Parquet tables using HDFS commands.

14. Run a simple count query on each table and compare performance.
 15. Explain the difference in how data is stored in TextFile, ORC, and Parquet formats.
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Part D: ACID and Transactional Tables

16. Create a transactional table for airlines_txn that supports INSERT, UPDATE, and DELETE operations.
 17. Insert data from the existing airlines table into airlines_txn.
 18. Update the active status of an airline to 'N' for a specific airline_id.
 19. Delete records of airlines that are inactive.
 20. Perform a major compaction on the transactional table and verify the results.
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Part E: Hive Functions and UDFs

21. Using built-in Hive functions, display the total number of airlines by country.
 22. Use a string function to extract the first three characters of the airport name for all airports.
 23. Use a conditional expression to classify airports based on altitude as 'High', 'Medium', or 'Low'.
 24. Create a Python-based UDF using the TRANSFORM clause to convert airline names to uppercase.
 25. Create another Python-based UDF to mask the first three letters of the airport name for privacy purposes.
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Part F: Analytical and Scenario-Based Queries

26. Find all airlines that operate direct flights (stops = 0) between two given airports.

27. Retrieve the total number of routes per airline and order the result by the highest number of routes.
 28. Identify countries where no active airlines are currently operating.
 29. Display the number of airports per timezone.
 30. Using a join across all three tables, display the airline name, source airport, destination airport, and number of stops for flights operated by airlines based in India.
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Part G: Conceptual and Descriptive Questions

31. Explain the purpose of bucketing in Hive and its importance in ACID tables.
32. Differentiate between managed and external tables in Hive with respect to data storage and deletion behavior.
33. Describe the Map-Side and Reduce-Side join process in Hive.
34. Explain how Hive determines whether to use a map-side join automatically.
35. Discuss the internal storage structure of ORC and Parquet files and how it helps improve query performance.